Syllabus

Course code	ECE 650 (graduate)	Course name	Advanced Computer Architecture	credits	3
Instructor	Name : Hyokeun Lee		Homepage : https://relacslab.github.io/		
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	Office hour : appointment is recommended				
1. Goals	Well defining an architecture is a key to achieve a robust and high-performance computing system. This lecture will take you to the road on various advanced technologies of computer architecture (e.g. out-of-order, prediction, coherence...). Although these technologies will be taught based on general-purpose processors, they have been ubiquitously extended in versatile platforms.				
2. Textbooks	**Main**: Patterson and Hennessy, "Computer Architecture: A Quantitative Approach" **Sub**: (1) A. Gonzalez et al., "Processor Microarchitecture: An Implementation Perspective" (2) Research papers included in class slides				
3. Prerequisites	▪ Computer Organization and Architecture (or similar class in undergraduates) ▪ C/C++ programming related classes				

4. Ratings (%)	Attendance	Homework	Mid-term	Final-term	Project	Others	Overall
	5	0	35	35	25	0	100

5. Agenda	Week	Contents
	1	Introduction to Computer Architecture
	2	Recap of ISA and Simple Microarchitecture
	3	Out-of-Order Microarchitecture (1)
	4	Out-of-Order Microarchitecture (2)
	5	Out-of-Order Microarchitecture (3)
	6	Branch Prediction
	7	Cache and Data Prefetch
	8	Mid-Term Exam
	9	Shared Memory: Cache Coherence (1)
	10	Shared Memory: Cache Coherence (2)
	11	Shared Memory: Memory Consistency (1)
	12	Shared Memory: Memory Consistency (2)
	13	Virtual Memory: Improving PTW Performance
	14	Virtual Memory: Enhancements for Various Domains
	15	Final-Term Exam

6. Notes for students	▪ F will be given if cheating is caught no matter what case is ▪ One grade lower if not taking either mid-term or final-term exam